PROBLEM STATEMENT NO. 7 | IBM GRANITE · RAG · MULTILINGUAL

AI Agent for Digital Financial Literacy

Project Report — RAG-Powered Financial Guidance for India | IBM Cloud Lite Services

LLM ENGINE **IBM Granite 4 H Small**	PLATFORM **IBM watsonx.ai (Lite Tier)**
LANGUAGES SUPPORTED **9 Indian Languages**	KNOWLEDGE DOMAINS **7 Financial Categories**
API FRAMEWORK **FastAPI + uvicorn**	VECTOR SEARCH **Milvus / FAISS (fallback)**

1. Problem Statement

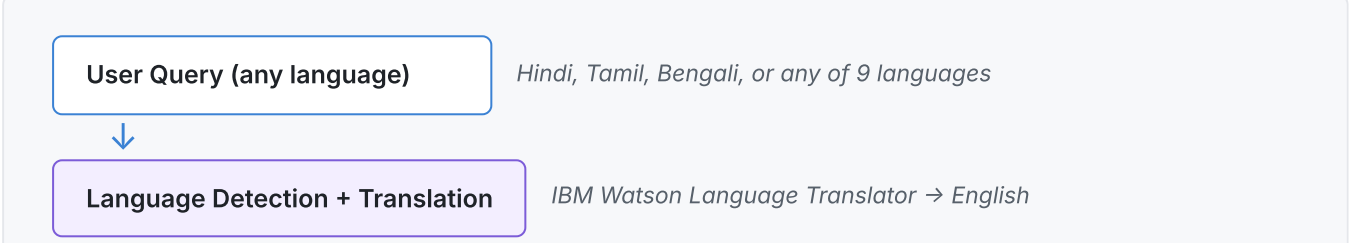
India has over 850 million internet users, yet financial literacy remains critically low — leaving millions vulnerable to UPI fraud, predatory loans, and poor savings decisions. The challenge is to build an **AI Agent for Digital Financial Literacy** that:

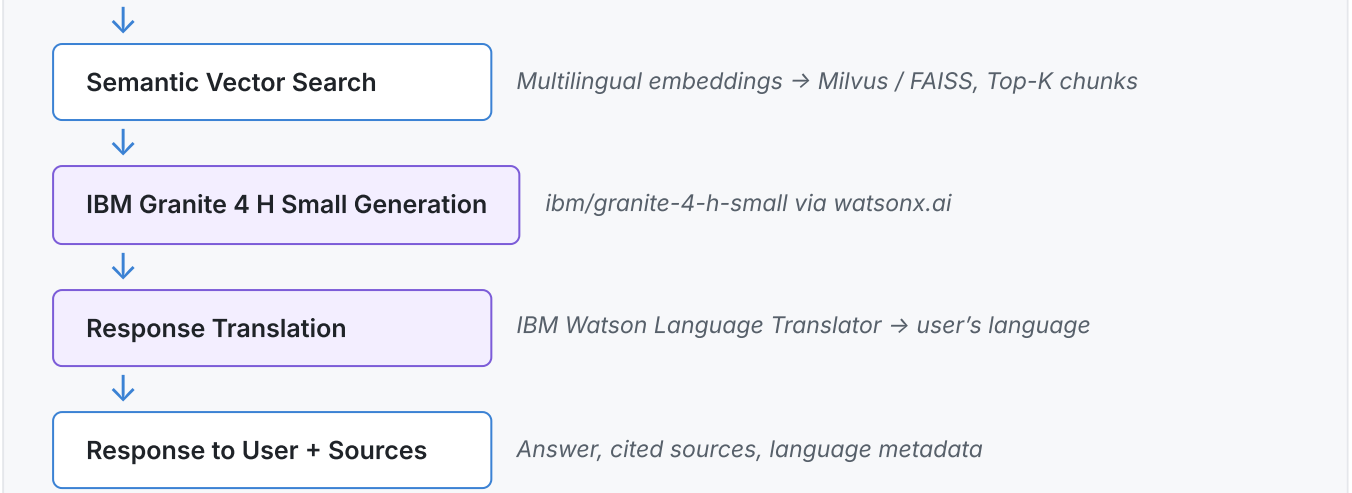
- Answers common financial questions in the user's native language
- Retrieves verified information from government portals (RBI, NPCI, SEBI)
- Protects users from online scams and fraud
- Guides users through UPI, EMI calculations, budgeting, and government schemes
- Is inclusive, accessible, and culturally relevant across Indian demographics

The solution must use **IBM Cloud Lite services and IBM Granite** as the mandatory technology stack.

2. Solution Architecture

The agent implements a **Retrieval-Augmented Generation (RAG)** pipeline with multilingual support at both input and output stages.





3. IBM Cloud Services (Mandatory)

SERVICE	ROLE IN SYSTEM	TIER	KEY CONFIG
IBM watsonx.ai	LLM inference via IBM Granite 4 H Small; generates financial answers from retrieved context	Lite	`WATSONX_API_KEY`, `WATSONX_PROJECT_ID`
IBM Watson Language Translator	Detects input language; translates query to English for retrieval; translates answer back to user language	Lite (1M chars/mo)	`LANGUAGE_TRANSLATOR_API_KEY`
IBM Cloud Object Storage	Persists raw PDF/HTML knowledge documents (RBI, NPCI, SEBI sources) for periodic re-indexing	Lite (25 GB)	`COS_API_KEY`, `COS_BUCKET_NAME`
IBM watsonx.data (Milvus)	Production-grade vector store for semantic retrieval; falls back to in-process FAISS for local dev	Lite	`MILVUS_HOST`, `MILVUS_PORT`

4. Project File Structure

```
# Project root
.env                # IBM API keys, model IDs, service URLs (never committed)
requirements.txt    # All Python dependencies with pinned versions
config.py           # Settings dataclass — reads from .env via python-dotenv
main.py             # FastAPI app: /chat, /ingest, /health, /languages endpoints
rag_engine.py       # Core RAG pipeline: detect → retrieve → Granite → translate
vector_store.py     # Milvus/FAISS wrapper: add_documents(), search()
translator.py       # IBM Watson Language Translator: detect, to_english, from_english
ingest.py           # 13 seed documents: RBI, NPCI, SEBI, AMFI, PMJDY knowledge
```

```
README.md           # Setup, API examples, architecture diagram
logs/agent.log      # Rotating application log (loguru, max 10 MB)
```

5. Key Module Descriptions

rag_engine.py — Core RAG Pipeline

Implements the **GraniteLLM** wrapper (thin client over `ibm_watsonx_ai.ModelInference`) and **RAGEngine.answer()**. The Granite model is called with greedy decoding (temperature 0.3, max 512 new tokens) and a carefully crafted system prompt that instructs the model to be helpful, cite official sources (RBI, NPCI, DigiLocker), and admit uncertainty.

vector_store.py — Semantic Document Store

Uses `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` (384-dim embeddings, supports all 9 languages natively) to embed documents and queries. Stores vectors in **Milvus** (IBM watsonx.data) with inner-product similarity search, with a transparent fallback to **in-memory FAISS** for development.

translator.py — Multilingual Bridge

Uses IBM Watson Language Translator for production-grade translation. Falls back to `langdetect` for language identification when the IBM service is unavailable. Supports pairwise translation between English and all 9 configured Indian languages.

ingest.py — Curated Knowledge Base

Seeds the vector store with 13 hand-crafted documents drawn from verified government sources: NPCI UPI guidelines, RBI fraud advisories, SEBI investor education, AMFI mutual fund guidance, CERT-In phishing alerts, and PMJDY / APY scheme details.

6. Environment Configuration (.env)

All secrets and service endpoints are externalised in `.env`. The file is structured in labelled sections:

```
# IBM watsonx.ai — Granite LLM
WATSONX_API_KEY=your_watsonx_api_key_here
WATSONX_PROJECT_ID=your_watsonx_project_id_here
WATSONX_URL=https://us-south.ml.cloud.ibm.com
GRANITE_MODEL_ID=ibm/granite-4-h-small

# IBM Watson Language Translator
LANGUAGE_TRANSLATOR_API_KEY=your_language_translator_api_key_here
LANGUAGE_TRANSLATOR_URL=https://api.us-south.language-translator.watson.cloud.ibm.com/insta

# IBM Cloud Object Storage (Lite — 25 GB free)
COS_API_KEY=your_cos_api_key_here
```

```
COS_INSTANCE_CRN=crn:v1:bluemix:public:cloud-object-storage:global:a/...
COS_BUCKET_NAME=fin-literacy-docs

# RAG & retrieval
RAG_TOP_K=5
RAG_SIMILARITY_THRESHOLD=0.75

# Multilingual languages (ISO 639-1)
SUPPORTED_LANGUAGES=en,hi,ta,te,bn,mr,gu,kn,ml
```

7. Multilingual Support (9 Indian Languages)

en English	**hi** हिन्दी (Hindi)	**ta** தமிழ் (Tamil)
te తెలుగు (Telugu)	**bn** বাংলা (Bengali)	**mr** मराठी (Marathi)
gu ગુજરાતી (Gujarati)	**kn** ಕನ್ನಡ (Kannada)	**ml** മലയാളം (Malayalam)

8. Knowledge Domains & Data Sources

UPI & Digital Payments
NPCI/BHIM product overview, transaction limits, VPA verification, 24×7 real-time transfers

Scam & Fraud Prevention
RBI advisories, KYC fraud patterns, phishing detection, CERT-In guidelines, cybercrime.gov.in reporting

Loans & Interest Rates
Safe personal loan rate range (10%–18%), RBI lender registry check, EMI formula with worked example

Savings & Investments
FD rates 2024, DICGC insurance, senior citizen rates, premature withdrawal penalties

Budgeting
50-30-20 rule, emergency fund steps (3–6 months), liquid mutual funds, budget tracking apps

Credit Health
CIBIL score (300–900 scale), 750+ target, credit utilisation, annual free report, EMI discipline

Government Schemes
PMJDY zero-balance account, RuPay card, Atal Pension Yojana, DigiLocker, DBT benefits

Insurance & Safety Nets
PMJJBY life insurance, PMSBY accident cover, DICGC deposit guarantee up to ₹5 lakh

9. REST API Endpoints

METHOD	ENDPOINT	DESCRIPTION
POST	`/chat`	Ask a financial question in any supported language. Returns answer, sources, and detected language.
POST	`/ingest`	Admin endpoint to add new documents to the RAG knowledge base (background task).
GET	`/health`	Returns service status and active Granite model ID.
GET	`/languages`	Returns all supported language codes with native names.
GET	`/docs`	Auto-generated OpenAPI / Swagger UI (FastAPI built-in).

10. Example Agent Interactions

USER QUERY	LANGUAGE	TOPIC
UPI से पैसे कैसे भेजें?	Hindi	UPI Payments
What is a safe interest rate for a loan?	English	Loans
ஆன்லைன் மோசடியை எப்படி தவிர்ப்பது?	Tamil	Scam Prevention
আমার বাজেট কীভাবে তৈরি করব?	Bengali	Budgeting
CIBIL score ಹೇಗೆ ಸುಧಾರಿಸಬಹುದು?	Kannada	Credit Score
Jan Dhan खाते के क्या फायदे हैं?	Hindi	Govt. Schemes

11. Setup & Deployment

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Fill in IBM Cloud credentials in .env
#    → WATSONX_API_KEY, WATSONX_PROJECT_ID, LANGUAGE_TRANSLATOR_API_KEY

# 3. Seed the vector knowledge base
python ingest.py

# 4. Start the agent
uvicorn main:app --host 0.0.0.0 --port 8000 --reload

# 5. Access Swagger UI
open http://localhost:8000/docs
```

12. Complete Technology Stack

LAYER	TECHNOLOGY	VERSION
LLM	IBM Granite 4 H Small (ibm-watsonx-ai)	1.0.8
Translation	IBM Watson Language Translator (ibm-watson)	8.1.0
Vector DB	Milvus (pymilvus) / FAISS fallback (faiss-cpu)	2.4.3 / 1.8.0
Embeddings	sentence-transformers paraphrase-multilingual-MiniLM-L12-v2	3.0.0
API Framework	FastAPI + uvicorn	0.111.0 / 0.29.0
Lang Detection	langdetect (offline fallback)	1.0.9
Doc Processing	pypdf, python-docx, BeautifulSoup4	4.2.0 / 1.1.2 / 4.12.3
Logging	loguru (rotating file + console)	0.7.2
Config	python-dotenv + custom Settings class	1.0.1
Resilience	tenacity (retry on transient IBM API errors)	8.3.0

13. Key Design Decisions

DECISION	RATIONALE
Translate-to-English for retrieval	All seed documents are in English (from RBI, NPCI); translating the query to English before embedding ensures maximum retrieval accuracy across all input languages.
Greedy decoding (temp 0.3)	Financial information must be factual and deterministic. Low temperature reduces hallucination risk compared to creative/sampling modes.
Similarity threshold 0.75	Filters out low-relevance chunks — prevents the model from being misled by loosely related passages.
FAISS fallback	Allows the agent to run fully offline for demos and development without Milvus, with zero code changes.
Offline lang-detect fallback	langdetect runs locally so language identification works even if Watson Translator quota is exhausted.
IBM Cloud Lite services only	Entire stack runs within IBM Cloud free tier — watsonx.ai Lite, Watson Translator Lite (1M chars/mo), COS Lite (25 GB), ensuring zero cost for hackathon/demo use.

14. Social Impact & Inclusivity

The agent directly addresses the digital divide in financial literacy across India's 1.4 billion population:

- **Language inclusivity:** 9 languages cover over 95% of India's linguistic population
- **Fraud protection:** Real-time guidance on identifying KYC fraud, phishing, and screen-share scams
- **Empowerment:** Explains government schemes (PMJDY, APY) that millions may not know they qualify for

- **Accessible finance:** EMI calculators, budget frameworks, and credit tips with no jargon
- **Trust:** All answers are grounded in RBI, NPCI, SEBI, and AMFI-verified sources with citations
- **Zero cost access:** Runs on IBM Cloud Lite — no infrastructure cost barrier for NGOs or state governments

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Data Sources: RBI (rbi.org.in) · NPCI (npci.org.in) · SEBI (sebi.gov.in) · AMFI India (amfiindia.com) · CERT-In (cert-in.org.in) · PMJDY (pmjdy.gov.in) · NPS CRA / APY (npscra.nsdl.co.in) · CIBIL (cibil.com)

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